**Course code: EE305**

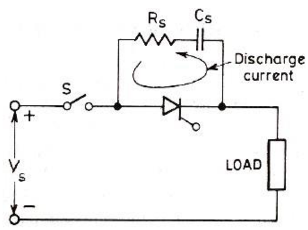
**Course Name: Power**

**Electronics**

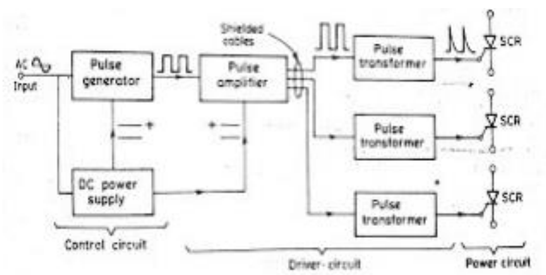
**PART – A**

1.

- With forward voltage across the anode and cathode of a thyristor, the two outer junctions areforward biased but the inner junction is reverse biased.
- This reverse biased junction act as a capacitor, with the entire anode to cathode voltage acrossit.
- Sudden rate of change of voltage, will lead to large charging current.
- This may damage the thyristor.
- Also this current plays the role of gate current and turns ON SCR, even though there is nogate signal (false triggering).
- Thus, rate of rise of forward voltage should be kept in specified limit.
- This can be achieved by connecting a capacitor in parallel with the thyristor.
- In a capacitor, voltage cannot be changed instantaneously.
- To limit the current during the discharging of the capacitor, a resistor is also connected inseries.




2.




Main features:

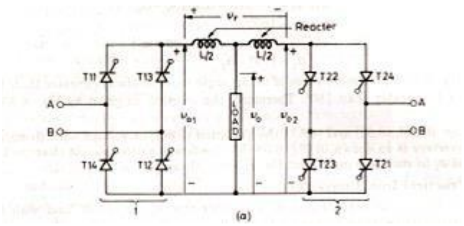
- If power circuit has more than one SCR firing circuit should produce gating pulse for each SCR at desired instant
- Firing circuit must contain a driver circuit

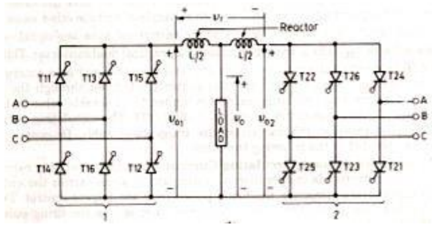
3.

- Circulating current is the current which is circulating through the converters.
- This current will not flow through the load.
- Circulating current will be present in the circuit, when both the converters are operatingsimultaneously.

Non-circulating current mode

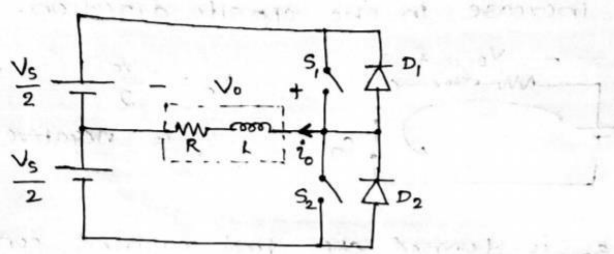
- In this mode of operation, only one converter is in operation at a time and it alone carries theentire load current.
- Only this converter receives the firing pulses from the trigger circuit.
- The other converter is blocked from conduction; this is achieved by removing the firingpulses from this converter.
- Thus only one converter is in operation and other converter is idle, there will not be anycirculating current.
- Hence this mode is called, non-circulating current mode.
- Circulating current mode
- In this mode of operation, both the converters are operating simultaneously.
- The firing pulses of the two converters are adjusted in such a way that $\alpha_1 + \alpha_2 = 180°$.
- By this way, average value of output voltage of both the converters can be made equal andthus possibility of DC circulating current can be avoided.
- Even though the average output voltages are equal, instantaneous values of output voltages ofthe two converters should not be equal.
- Thus, there will be ac circulating current.
- The amount of this circulating current can be limited by inserting a reactor in-between the converters 1 and 2.




4.

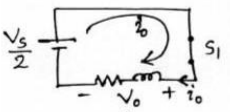
1-Phase half-bridge Inverter with RL-load




Due to the presence of inductance, load current can not be reversed Instantaneously. Thus switch $S_2$ should not be turned ON, at the same instant when $S_1$ is turned OFF.

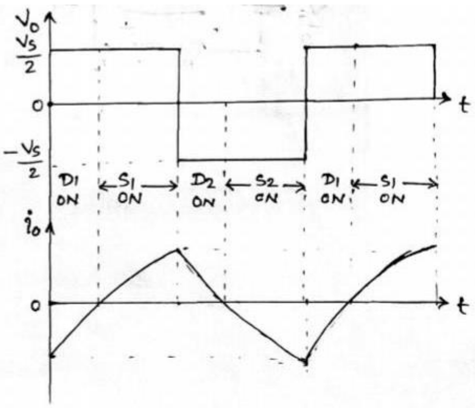
## Working

*) When $S_1$ is ON

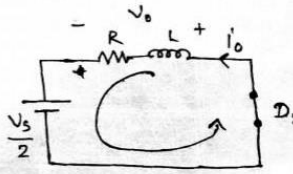



$V_o = \frac{V_s}{2}$ , $i_o$ is +ve.

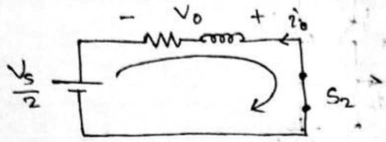
load current increases from zero and reaches a maximum value.




*) When $S_1$ is turned OFF, load current should not be reversed instantaneously. Thus ~~diode~~ current continue to flow in the same direction through $D_2$. Thus $V_o = -V_s/2$ and thus $i_o$ decreases. ⇒



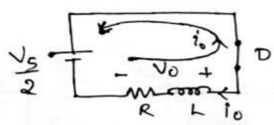

*) When $i_o$ decreases and reaches zero, Switch '$S_2$' is turned ON. Thus load current starts ~~to~~ to increase in the opposite direction.




$V_o = -\frac{V_s}{2}$

$i_o$ is negative.

*) When $S_2$ is turned OFF, load current continues to flow through the diode $D_1$. Thus Load current decreases and reaches zero.



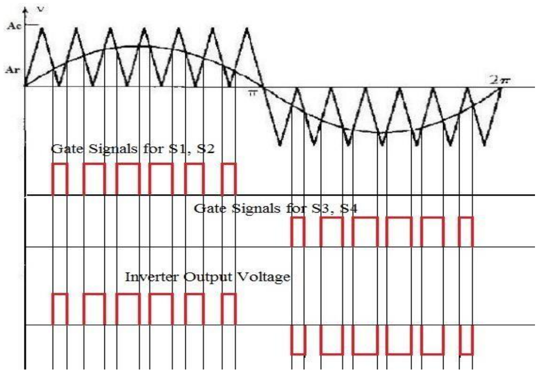

$V_o = \frac{V_s}{2}$

$i_o$ is negative.

5.

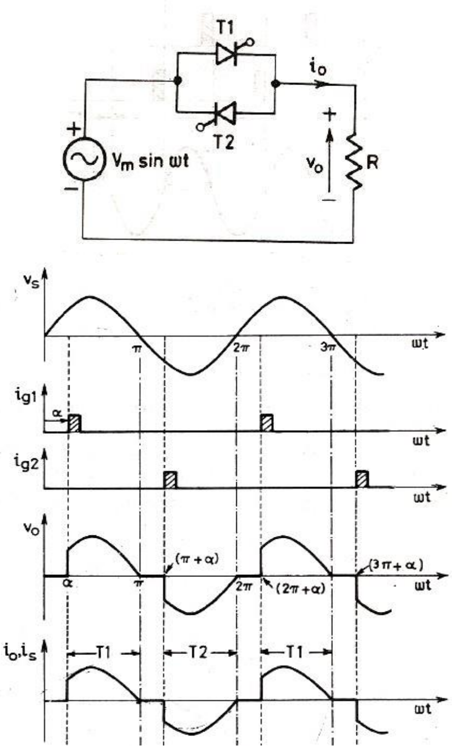
In this modulation, there are multiple number of pulses per half cycle and width of the pulses varies asa function of sine.

The gate signals are generated by comparing a sinusoidal reference signal ($V_r$) with a triangular carrier wave ($V_c$).



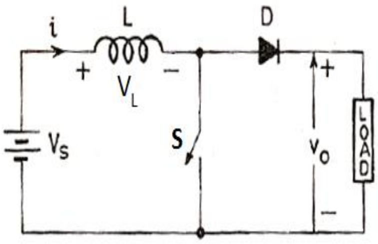
V
Ac
Ar
π
2π
Gate Signals for S1, S2
Gate Signals for S3, S4
Inverter Output Voltage

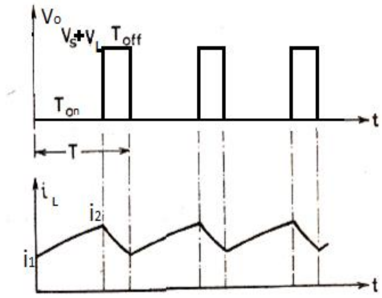

6.

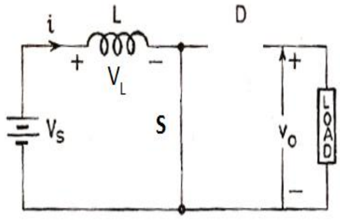


T1
T2
$i_o$
$V_m \sin \omega t$
$v_o$
R
$v_s$
$i_{g1}$
$i_{g2}$
$v_o$
$i_o, i_s$
α
π
2π
3π
$(\pi+\alpha)$
$(2\pi+\alpha)$
$(3\pi+\alpha)$
ωt

7.

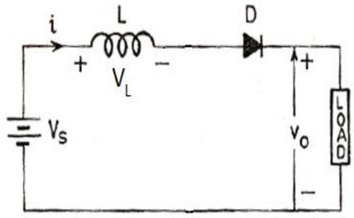



When switch S is ON, inductor L stores energy from the source through the switch. Thus inductor current increases from i1 to i2.

Thus, $V_L = V_S$




When switch S is turned OFF, inductor L releases its stored energy to the load through the diode. Thus inductor current decreases from i2 to i1.

Thus, $V_L = V_S - V_O$

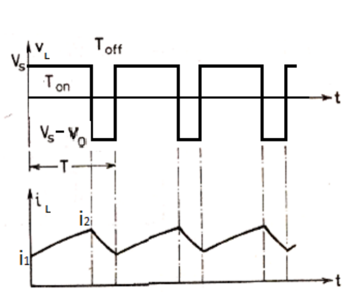
**Average Output voltage**

At steady state, average energy stored in an inductor will be equal to the energy released by it.

Or we can say that for an inductor sum of energy stored and released over a period will be zero.

Energy = $voltage \times average\ current \times time$

**Average value of current** = $\dfrac{i_1 \times T + \frac{1}{2} T \times (i_2 - i_1)}{T} = i_1 + \frac{i_2}{2} - \frac{i_1}{2} = \frac{i_1}{2} + \frac{i_2}{2} = \frac{(i_1 + i_2)}{2}$



Energy stored = $V_s \times \frac{(i_1 + i_2)}{2} \times T_{on}$

Energy released = $(V_s - V_o) \times \frac{(i_1 + i_2)}{2} \times T_{off}$

Applying volt-second balance,

$$V_s \times \frac{(i_1 + i_2)}{2} \times T_{on} + (V_s - V_o) \times \frac{(i_1 + i_2)}{2} \times T_{off} = 0$$

$$V_s \times T_{on} + (V_s - V_o) \times T_{off} = 0$$

$$V_s T_{on} + V_s T_{off} - V_o T_{off} = 0$$

$$V_s (T_{on} + T_{off}) - V_o T_{off} = 0$$

$$V_o = V_s \frac{(T_{on} + T_{off})}{T_{off}} = V_s \left( \frac{T_{on}}{T_{off}} + 1 \right)$$

$$V_o = V_s \left( \frac{T_{on}}{T - T_{on}} + 1 \right)$$

$$V_o = V_s \left( \frac{1}{\frac{T - T_{on}}{T_{on}}} + 1 \right) = V_s \left( \frac{1}{\frac{T}{T_{on}} - 1} + 1 \right)$$

$$V_o = V_s \left( \frac{1}{\frac{1}{D} - 1} + 1 \right) = V_s \left( \frac{D}{1 - D} + 1 \right)$$

$$V_o = V_s \left( \frac{D + 1 - D}{1 - D} \right) = \frac{V_s}{1 - D}$$

8.

$V_{in} = 15V$

$V_o = 5V$

$R = 1K\Omega$

$\Delta V_o = 15mV = 0.015V$

$f = 30KHz$

$\Delta i_L = 0.5A$

a.) $V_o = DV_{in}$

duty cycle, $D = \frac{V_o}{V_{in}} = \frac{5}{15} = 1/3$

$= \underline{0.333}$

b.) filter inductance, $L = \frac{V_o(1-D)}{f\Delta i_L}$

$= \frac{5(1-1/3)}{30\times10^3\times0.5}$

$= \underline{0.22\ mH}$

c.) filter capacitance, $C = \frac{V_o(1-D)}{8f^2 L\Delta V_o}$

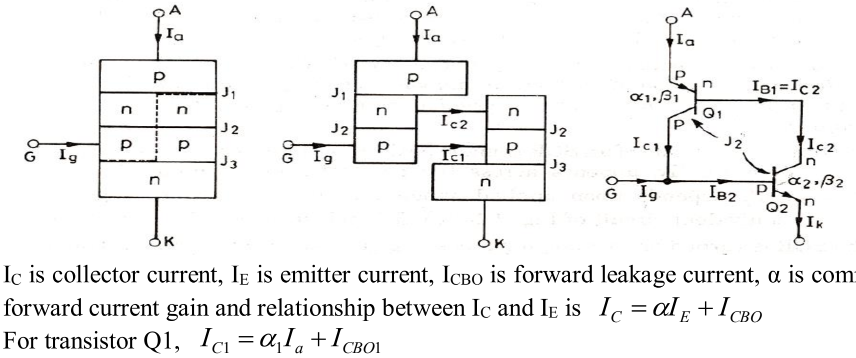
$= \frac{5(1-1/3)}{8\times(30\times10^3)^2\times0.22\times10^{-3}\times0.015}$

$= \underline{138.89\ \mu F}$

**PART – B**

9.

a)




- $I_C$ is collector current, $I_E$ is emitter current, $I_{CBO}$ is forward leakage current, α is common base forward current gain and relationship between $I_C$ and $I_E$ is $I_C = \alpha I_E + I_{CBO}$
- For transistor Q1, $I_{C1} = \alpha_1 I_a + I_{CBO1}$
- For transistor Q2, $I_{C2} = \alpha_2 I_k + I_{CBO2}$
- By applying KCL $I_a = I_{C1} + I_{C2}$

$$I_a = \alpha_1 I_a + I_{CBO1} + \alpha_2 I_k + I_{CBO2}$$
$$I_a = \alpha_1 I_a + I_{CBO1} + \alpha_2 (I_{B2} + I_{C2}) + I_{CBO2}$$
$$I_a = \alpha_1 I_a + I_{CBO1} + \alpha_2 (I_g + I_{C1} + I_{C2}) + I_{CBO2}$$
$$I_a = \alpha_1 I_a + I_{CBO1} + \alpha_2 (I_g + I_a) + I_{CBO2}$$
$$I_a = \frac{\alpha_2 I_g + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)}$$

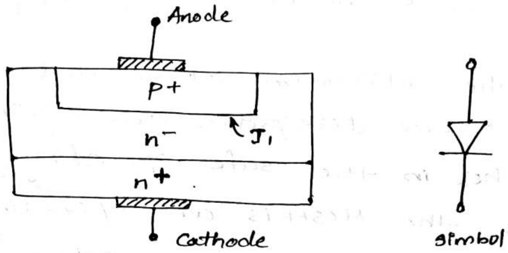
- During forward blocking mode, only leakage current will flow through SCR and $I_g = 0$.
- Since anode current is small, α1 and α2 will be small
- Thus SCR remains in OFF state.
- When $I_g$ is applied, Ia increases and thus α1 and α2 will increase.
- Again Ia increases.
- This cumulative process will turn on the SCR.

b)

★ POWER DIODES

Structure

Power diodes differ in structure from signal diodes. A signal diode constitute a simple p-n junction. But in power diode, it consists of heavily doped n+ subtrate. on this subtrate n− layer is epitaxially grown. Then a heavily doped p+ layer is diffused into n− layer to form the anode of power diode.




structure.

This shows that n− layer is the basic structural feature not found in signal diodes.

The function of n− layer is to absorb the depletion layer of the reverse biased $p^+n^-$ junction $J_1$.

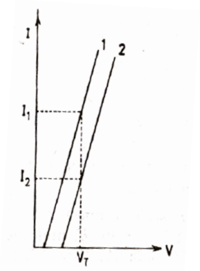
The breakdown voltage needed in a power diode, governs the thickness of n− layer; greater the breakdown voltage, more the n− layer thickness.

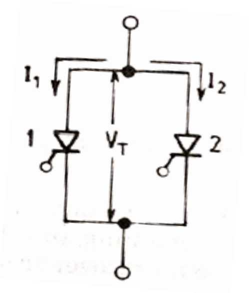
The drawback of n− layer is to add significant ohmic ~~loss~~ resistance to the diode when it is conducting a forward current. This leads to lage

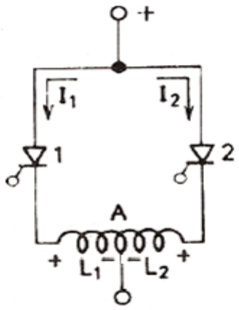
10.

a)

- When current required by the load is more than the rated current of a single SCR, two or more SCRs are connected in parallel.



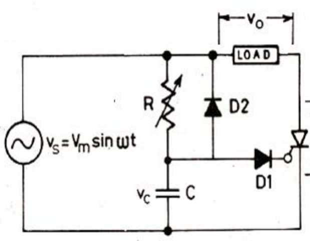



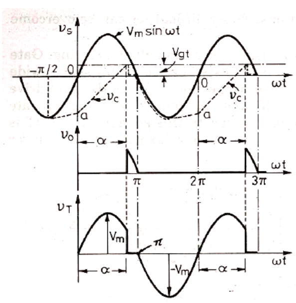



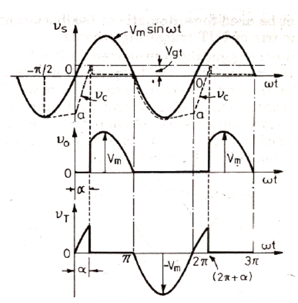

- Current distribution can be made more uniform by the magnetic coupling of the parallel paths as shown.
- A is the mid-point of the reactor.
- If $I_1 = I_2$, Flux produced by 2 halves of the reactor cancel each other.
- If currents are not equal, the flux linkages induce emfs in $L_1$ and $L_2$.
- Emf across $L_1$ opposes the flow of $I_1$ whereas that across $L_2$ aids the flow of $I_2$.
- Thus minimize the unbalance of currents in the parallel paths

b)

- Limited range of firing angle control can be overcome by RC firing circuit.
- Firing angle can be controlled between 0 and 180.
- D1 is used to prevent negative voltage across gate-cathode junction.
- During the -ve half cycle, capacitor charges from the source through the diode D2, with lower plate positive.
- During the +ve half cycle, D2 discharges through R and value of R determines the discharging time.
- Once the capacitor voltage reaches zero, it will charge in the opposite direction, with upper plate positive.
- Whenever, the capacitor voltage equal to the required gate trigger voltage ($V_{gt}$), SCR turns ON



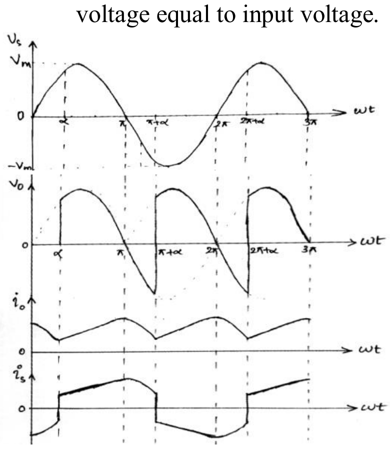



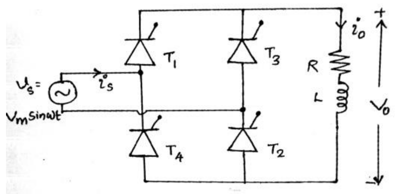




11.

- During the positive half cycle, SCR $T_1$ and $T_2$ will be in forward biased condition whereas $T_3$ and $T_4$ are reverse biased.

- Whenever, the firing pulses are given to $T_1$ and $T_2$, they will turn ON and act as closed switches.
- Here, at ωt=α, firing signals are given to gate terminals of $T_1$ and $T_2$. Thus SCRs turns ON and output voltage will be equal to input voltage.
- Since the load is RL, current can not be changed instantaneously and also it lags the voltage.
- Thus, current starts increasing from zero and reaches a maximum value at ωt=π. That is, at ωt=π, output voltage become equal to zero but current will not be zero.
- That is, current through SCRs will not be zero, thus SCRs continue in conduction with output voltage equal to input voltage.




- After ωt=π, since the output voltage is negative, output current starts to decrease.
- If the value of inductor is considerably large, the time constant will be high and more time will be needed for charging and discharging of inductor.
- Thus, output current will not become zero upto ωt= π+α.
- But, at ωt= π+α, firing signals are given to gate terminals of $T_3$ and $T_4$.
- Thus $T_3$ and $T_4$ turn ON and output current will continue through them.
- Since the current got separate path, $T_1$ and $T_2$ will get turned OFF.
- Since $T_3$ and $T_4$ are ON, output voltage will be equal to the magnitude of input voltage but opposite in polarity.
- Output current starts increasing and reaches a maximum value at ωt=2π.
- That is, at ωt=2π, output voltage become equal to zero but current will not be zero. Thus output voltage become negative after 2π.
- Thus output current starts decreasing and reaches a minimum value at ωt=2π+α.
- At ωt=2π+α, again firing signals are given to $T_1$ and $T_2$, and the process continues.

b)

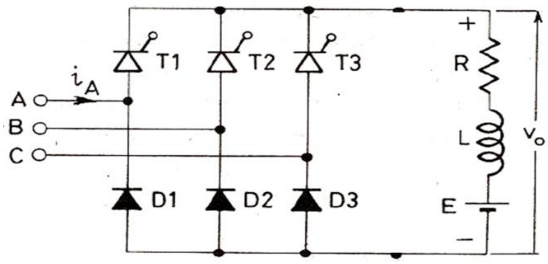
$$V_o = \frac{\int_{\alpha}^{\pi+\alpha} (V_m \sin \omega t) d\omega t}{\pi} = \frac{V_m}{\pi} \int_{\alpha}^{\pi+\alpha} (\sin \omega t) d\omega t$$

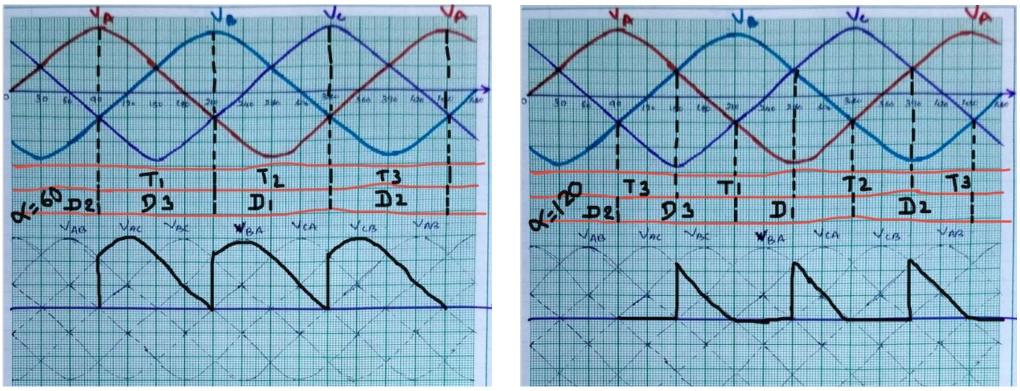
$$= \frac{V_m}{\pi} [-\cos \omega t]_{\alpha}^{\pi+\alpha} = \frac{V_m}{\pi} [(-\cos(\pi+\alpha)) - (-\cos\alpha)]$$

$$= \frac{V_m}{\pi} [(\cos\alpha) - (-\cos\alpha)] = \frac{V_m}{\pi} (2\cos\alpha)$$

**PART – C**

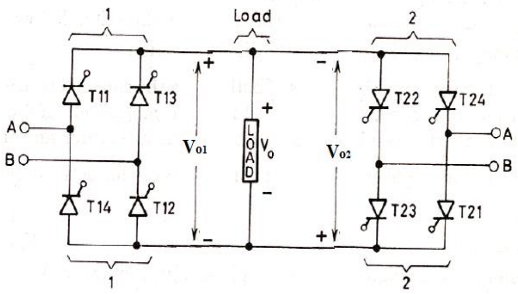
12.

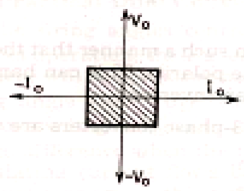



13.

a)




**<u>Quadrant – 1</u>**

Converter-1 operates with $\alpha_1$ less than 90°.

Thus current and voltage will be positive.

**<u>Quadrant – 2</u>**

Converter-2 operates with $\alpha_2$ greater than 90°.

Thus current will be negative and voltage will be positive.

**<u>Quadrant – 3</u>**

Converter-2 operates with $\alpha_2$ less than 90°.

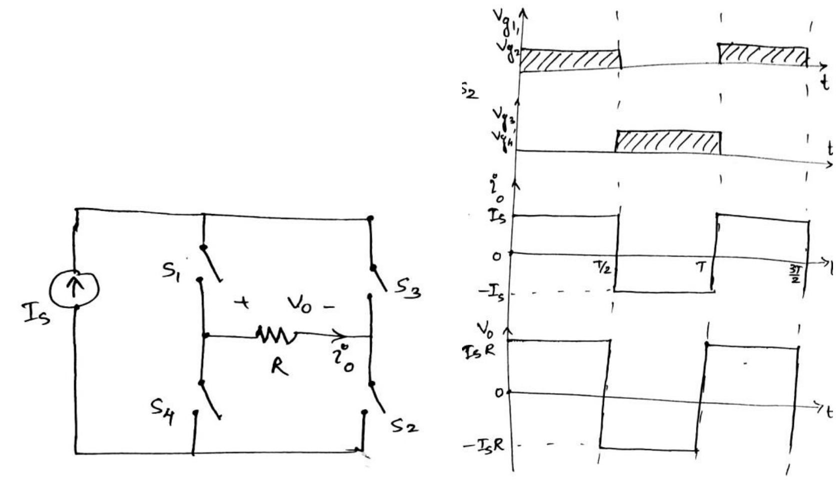
Thus current and voltage will be negative.

**<u>Quadrant – 4</u>**

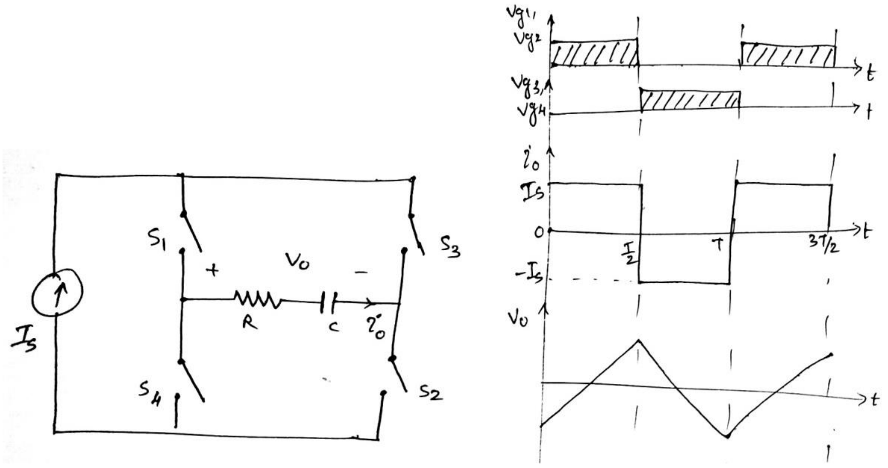
Converter-1 operates with $\alpha_1$ greater than 90°.

Thus current will be positive and voltage will be negative.

b)

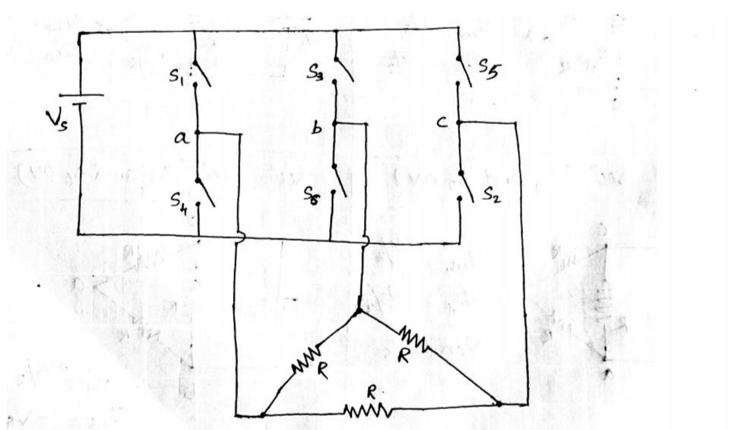


$I_s$
$S_1$
$S_3$
$S_4$
$S_2$
+ $V_0$ −
R
$i_0$
$V_{g1}$, $V_{g2}$
$V_{g3}$, $V_{g4}$
$i_0$
$I_s$
0
$-I_s$
T/2
T
3T/2
$V_0$
$I_s R$
$-I_s R$
t

$I_s$
$S_1$
$S_3$
$S_4$
$S_2$
+ $V_0$ −
R
c
$i_0$
$V_{g1}$, $V_{g2}$
$V_{g3}$, $V_{g4}$
$i_0$
$I_s$
0
$-I_s$
T/2
T
3T/2
$V_0$
t


14.



$V_s$
$S_1$
$S_3$
$S_5$
a
b
c
$S_4$
$S_6$
$S_2$
R
R
R

## ⇒ $120^\circ$ Conduction Mode.

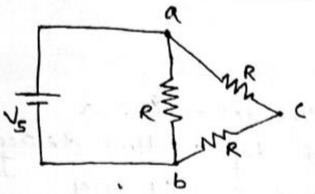
In this operating mode, each device conducts for a duration of $120^\circ$. Devices are triggered by a time delay of $60^\circ$ in the order $S_1, S_2, S_3, S_4, S_5$ and $S_6$. At any instant of time, 2 devices will be conducting.

From the switching waveforms, it is clear that,

$0^\circ - 60^\circ \Rightarrow S_6$ and $S_1$ are ON
$60^\circ - 120^\circ \Rightarrow S_1$ and $S_2$ are ON
$120^\circ - 180^\circ \Rightarrow S_2$ and $S_3$ are ON
$180^\circ - 240^\circ \Rightarrow S_3$ and $S_4$ are ON
$240^\circ - 300^\circ \Rightarrow S_4$ and $S_5$ are ON
$300^\circ - 360^\circ \Rightarrow S_5$ and $S_6$ are ON

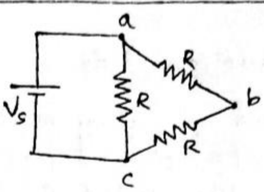
and this sequence continues.

### $0^\circ - 60^\circ$ ($S_6$ and $S_1$ ON)



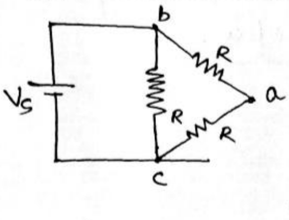

$V_{ab} = V_s$
$V_{bc} = -\frac{V_s}{2}$
$V_{ca} = -\frac{V_s}{2}$

### $60^\circ - 120^\circ$ ($S_1$ and $S_2$ ON)



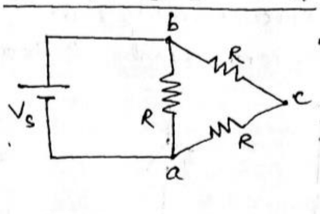

$V_{ab} = V_s/2$
$V_{bc} = V_s/2$
$V_{ca} = -V_s$

### $120^\circ - 180^\circ$ ($S_2$ and $S_3$ ON)



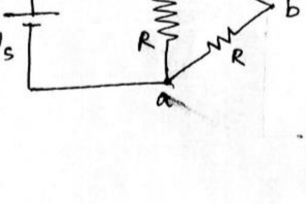

$V_{ab} = -V_s/2$
$V_{bc} = V_s$
$V_{ca} = -V_s/2$

### $180^\circ - 240^\circ$ ($S_3$ and $S_4$ ON)



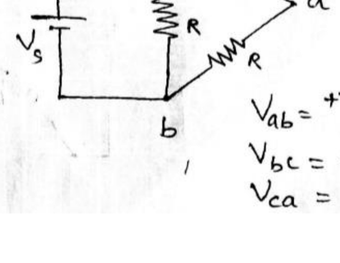

$V_{ab} = -V_s$
$V_{bc} = V_s/2$
$V_{ca} = V_s/2$

### $240^\circ - 300^\circ$ ($S_4$ and $S_5$ ON)

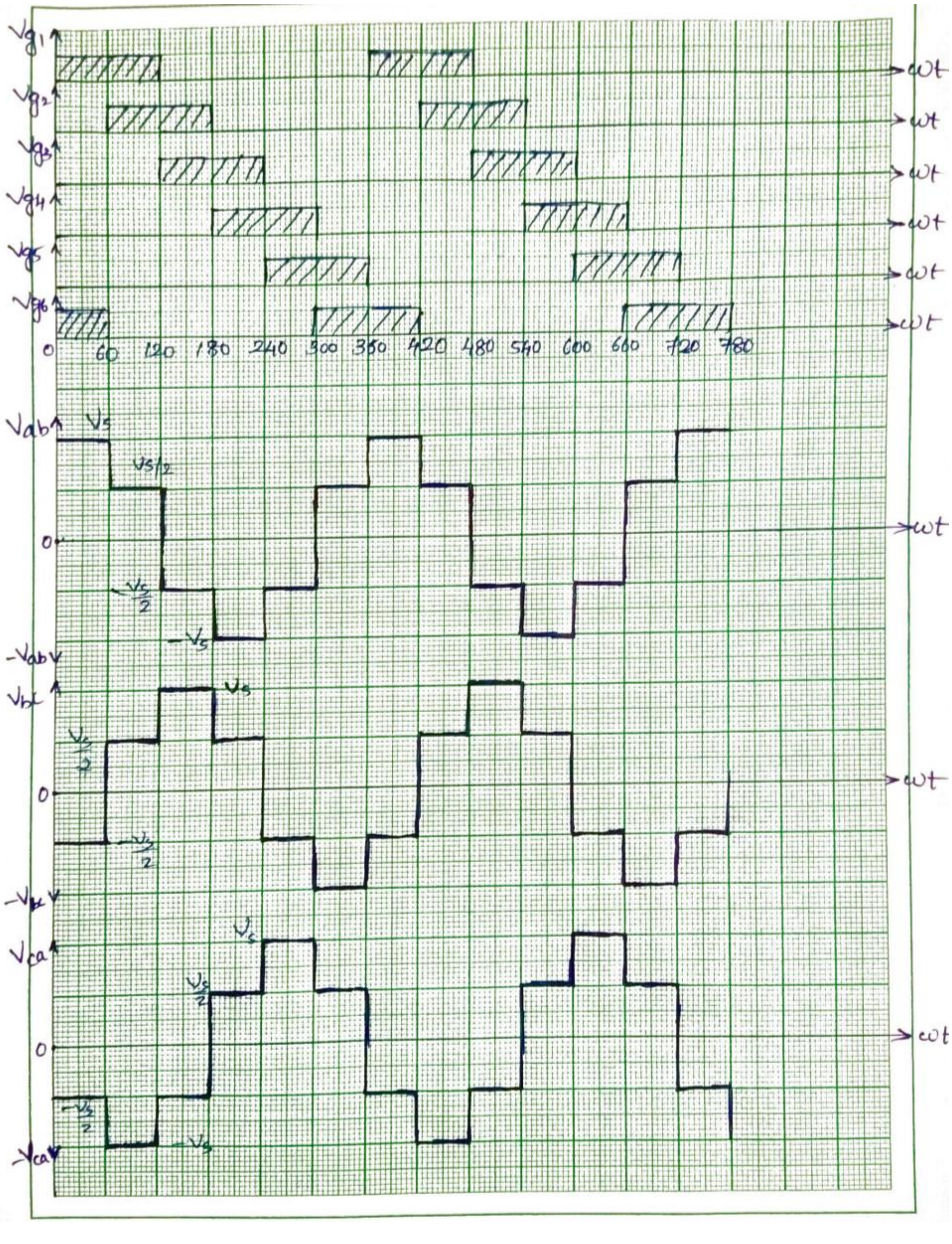



$V_{ab} = -V_s/2$
$V_{bc} = -V_s/2$
$V_{ca} = V_s$

### $300^\circ - 360^\circ$ ($S_5$ and $S_6$ ON)




$V_{ab} = +V_s/2$
$V_{bc} = -V_s$
$V_{ca} = V_s/2$



Vg1
Vg2
Vg3
Vg4
Vg5
Vg6
ωt
0
60
120
180
240
300
360
420
480
540
600
660
720
780
Vab
Vs
Vs/2
-Vs/2
-Vs
-Vab
Vbc
-Vbc
Vca
-Vca

**PART – D**

15.

a)

The technique of changing the width of switching pulses given to the switching devices in the inverter to control the output voltage is called **Pulse Width Modulation (PWM) Technique**.

**Amplitude Modulation Index**: It is defined as the ratio of the amplitude of reference signal to the amplitude of carrier signal

Amplitude Modulation Index $= \frac{A_r}{A_c}$

**Frequency Modulation Index**: It is defined as the ratio of frequency of carrier signal to the frequency of reference signal.

Frequency Modulation Index $= \frac{f_c}{f_r}$

b)

$$V_{o,rms} = \frac{V_m}{\sqrt{2\pi}} \sqrt{(\pi-\alpha) + \frac{\sin 2\alpha}{2}}$$

$\alpha = 90^\circ$

$$V_{o,rms} = \frac{230\times\sqrt{2}}{\sqrt{2\pi}} \sqrt{\left(\pi-\frac{\pi}{2}\right) + \frac{\sin 180}{2}}$$

$$= \frac{230\sqrt{2}}{\sqrt{2}\cdot\sqrt{\pi}} \sqrt{\pi/2}$$

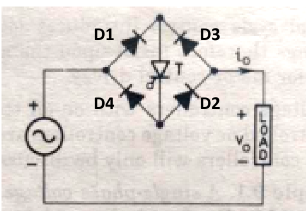
$$= \frac{230}{\sqrt{2}} = \underline{162.635\ V}$$

$$\text{Power factor} = \frac{\text{RMS output voltage}}{\text{Rms input voltage}}$$

$$= \frac{162.635}{230} = \underline{0.707}$$

16.

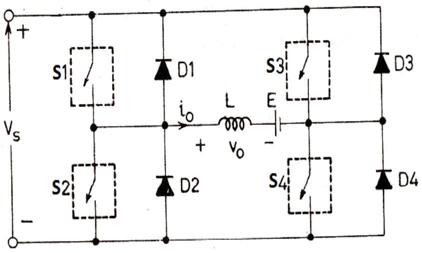
a)

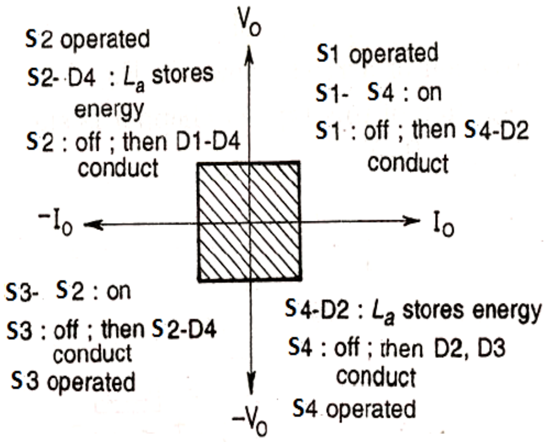



Here thyristor T is able to operate during positive as well as negative half cycles. During positive half cycle, D1, D2 and T will be forward biased and thus current can flow through this path to load. During negative half cycle, D3 and D4 will be for ward biased and current can flow to load through the thyristor T.

Advantage of this circuit as compared with two thyristor ACVC is that there is no isolation issues.

b)

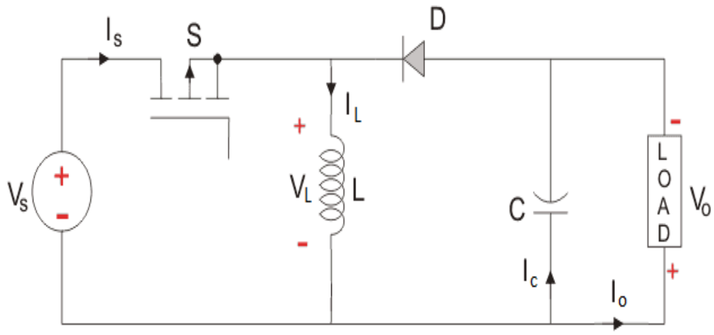



17.

a)

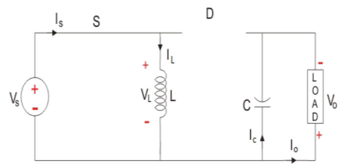
- Buck-boost converter is a voltage step-down step-up converter.
- Output voltage can be less than or greater than the input voltage.




→ When switch S is ON, diode D will be reverse biased and it will not conduct.

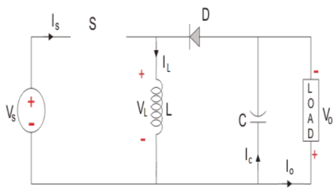
→ Voltage across the inductor, $V_L = V_s$

→ Since the voltage across the Inductor is positive, it stores energy and inductor current $I_L$ will increase from a minimum value $I_1$ to a maximum value $I_2$.




→ During this time, capacitor discharges and provide the output current.

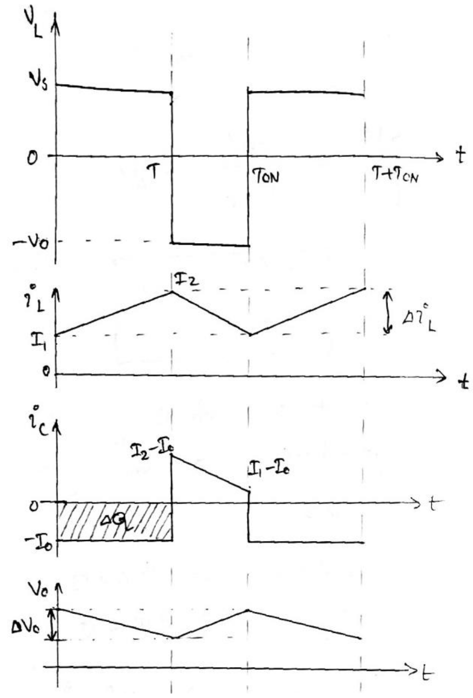
→ Thus, $I_o = -I_c$




→ When switch S is turned OFF, inductor current finds a path through the diode D.

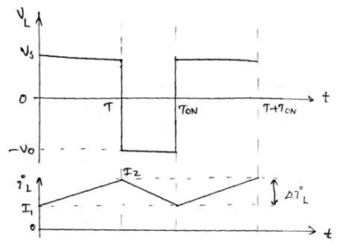
→ Voltage across the inductor, $V_L = - V_o$

→ Since $V_L$ is negative, inductor current $I_L$ decreases from $I_2$ to $I_1$.




b)

**Design of Inductor**

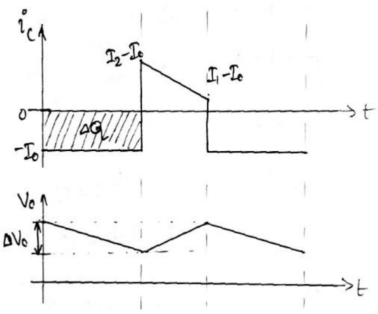



We have $V_L = L\dfrac{di_L}{dt}$

During $T_{ON}$ $\quad V_s = L\dfrac{\Delta i_L}{T_{ON}}$

$$L = V_s\frac{T_{ON}}{\Delta i_L} = V_s\frac{DT}{\Delta i_L} = \frac{DV_s}{f\Delta i_L}$$

**Design of Capacitor**




For a capacitor, $\quad Q = CV \ or \ \Delta Q = C\Delta V$

Charge stored during the time of capacitor charging = area of current vs time curve during charging. $\quad \Delta Q = I_o T_{ON}$

During the charging of capacitor, change in voltage is $\quad \Delta V_o$

Thus, $\quad C\Delta V_o = I_o T_{ON}$

$$C = \frac{I_o T_{ON}}{\Delta V_o} = \frac{I_o DT}{\Delta V_o}$$

$$C = \frac{DI_o}{f\Delta V_o}$$